cloudtorch / develop — Step 5a session summary: from velocity collapse to a full-cube fit

2026-08-25 — diagnosed why 5a's velocities collapsed, fixed it with a single reg backstop knob, mapped where the fit spoils, scaled 5a from the 64^2 laptop test to the *full* MiHI cube, and added scripted twins + movies.

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1. NOTEBOOK 5A — CURRENT STATUS

Scaled from the laptop crop to the **full bordered cube**: `N_T`=32 frames × 116×130 px = **482,560** samples, `N_FREQ`=128, **minibatch** `BATCH`=16384, `LR`=1e-2, 3000 iters (∼116 s on one GPU). Regularisation is a gentle **backstop** (`W_REG`=1e-4 over re-tuned per-group weights); `log a` frozen. **Result:** mean per-pixel $\chi^2 = \mathbf{0.233}$, RMS **3.67%**, with physically *structured, sign-crossing* velocities (cloud 1 red $-9/+11/+74$, cloud 2 blue $-105/-23/+1.5$ km/s). The notebook also dumps per-frame PNGs to `../figs/debug/` and writes `fits_and_params.npz`. This is the first **full-FOV, all-frames** given-background fit — healthy, converged, physical.

2. ADVANCEMENTS

- **Root-caused the collapse** (clean 2×2 factorial): the *explicit regularisation* — not Fourier bandwidth, init, or frame window — flattens the weakly-constrained velocity field. `docs/step_5a_factorial.pdf`.
- **One global backstop knob** `W_REG` scaling all reg weights: `W_REG`=1e-3 reproduces 4a on the laptop window (χ^2 0.181 vs 4a 0.170), *verified* by a full notebook re-run.
- **Parameter search** `step_5a_reg_scan.py`: mapped the spoil transition (below).
- **Scale-up** to the full cube with minibatch (above).
- **Tooling:** `step_6a_reg_fit.py` (config-driven scripted twin of 5a, `fit_cube.py` conventions: `-config/-set/-dump-config`, logs `config_used.json`); `png_to_video.py` and `npz_inversion_videos.py` (fixed-scale fit & parameter movies).

3. PERFORMANCE

stage	window	reg	$\langle\chi^2\rangle$	RMS	velocities
4a reference	$64^2 \times 4$, 1.2k	off	0.170	3.25%	structured, *runaway tails*
5a reproduce 4a	$64^2 \times 8$, 10k	`W_REG` 1e-3	0.181	3.36%	structured, tails clipped
5a full cube	116·130·32, 3k mb	`W_REG` 1e-4	**0.233**	3.67%	sign-crossing

`W_REG` scan (4k iters, laptop window): χ^2 rises *monotonically* 0.16→0.21→0.25→0.29→0.38→0.57 across `W_REG`=0, 1e-3, 3e-3, 1e-2, 1e-1, 1. Usable backstop ≤**1e-3**; **knee** ∼**3e-3**; clearly spoiled ≥**1e-2**. `vlos2` (weakest-constrained) collapses first, the moment reg turns on.

4. LESSONS LEARNED

- **"Small in loss" ≠ "harmless."** A 1–3% penalty dominates a weakly-constrained (near-null) direction: the fixed background owns the line wings, so the data barely pulls on `vlos`, and any appreciable smoothness/hinge flattens it. Regularise weakly-constrained parameters *only* as a backstop.
- **No strong-reg sweet spot here:** reg trades ∼monotonically against *both* χ^2 and velocity structure. The fix is a gentler penalty, or a robust/far-outer hinge that clips outliers *without* flattening.
- **Activation design is physics.** Identity for signed/unbounded `vlos` + a soft outer hinge is correct; bounded transforms (`S` sigmoid, `dtau/dv` softplus) only where a real physical bound exists; `SiLU` hidden units stay (smoothness reg needs a C^∞ double-backward — `ReLU` would be degenerate).
- **Architecture:** prefer *wide* over *deep* for a Fourier-feature field — the random features already supply the frequencies, so depth (no residuals here) mostly costs conditioning.
- **Optimisation:** the explicit autograd reg slows convergence; **minibatch** + **higher** `LR` makes the full 482k-pixel cube tractable in ∼2 min.

5. NEXT

- Replace the velocity smoothing with a **robust loss / far-outer hinge** (clip tails, keep structure), then re-scan `W_REG` on the *full* cube.
- Sub-factorial (`w_t`-only vs hinge-only) to pin the dominant collapse term; carry the tuned config into `step_6a_reg_fit.py` / `fit_cube.py` for deployment.

Artefacts this session: `step_5a_reg_scan.py`, `step_6a_reg_fit.py`, `png_to_video.py`, `npz_inversion_videos.py` (in `figs/debug/`), `docs/step_5a_factorial.pdf`. Note `composite_model.py` `_CLOUD_INIT` fallback was changed to an asymmetric $-50/0$ experiment (unrelated to the pinn init, which stays ∓ 20/scale 20).